

Optimizing School Entrance Throughput using Automated Multi-Sensor Classification to Eliminate Manual Bottlenecks.

Problem Scenario
Our school entrance is consistently slow and congested because security guards perform manual bag checks on every student. This creates a safety risk due to crowd crushing and wastes instructional time. The Objective: To transition from manual checks to an automated, exception-based system that uses multi-variable sensors to distinguish between harmless personal items and prohibited threats (weapons/vapes) without stopping the flow of students.

School Metal Detector
School metal detectors are intended to enhance safety by identifying prohibited metal objects; however, their effectiveness is limited by interacting signal constraints. Variations in signal amplitude and spatial density make it difficult to accurately determine how much metal is present and whether it represents a small harmless object or a larger potential threat. Multiple small metallic items carried close together can produce signal strengths similar to those of dangerous objects, leading to false alarms or missed detections. At the same time, phase angle and signal duration constraints affect the detector's ability to correctly classify detected metals. Similar phase responses among different metal types, combined with variations in how long a signal remains above the detection threshold due to walking speed or object placement, reduce the system's reliability.

ITERATION 1 - Large Metal Object Checking	
Problem Identification	How can we automate the detection of medium-sized and large-sized metal objects at the school entrance, ignoring small and non-threatening metal objects?
Decomposition	To determine if an object is metal and what size it is, we break the signal into two metrics: 1. Signal Amplitude Threshold (Is it metal? How much?) - o Logic: A "Noise Floor" established to ignore tiny metal items like buttons or zippers. - o Size Scaling: Higher amplitude typically correlates to a larger metallic mass, allowing the system to categorize the object as "Small" (ignore), "Medium" (vape/phone", or "Large" (potential weapon). 2. Signal Duration Threshold (The Speed/ Size Cross-Check):

	○ This measures the length of time the signal stays above the threshold as the student moves through the gate. ○ Logic: A large object moving at a normal walking speed will create a longer duration than a small object moving at the same speed. ○ Validation: If high amplitude is paired with a specific duration, it confirms a "solid mass" rather than a momentary interference spike.
Pattern Recognition	○ The "Mass" Signature: Recognizing that high-density metallic objects create a distinct peak in signal amplitude that stays consistent across different passes. ○ The "Clutter" Pattern: Low Amplitude + Short Duration = Harmless items (coins, keys). ○ The "Threat" Pattern: High Amplitude + Sustained Duration = Large objects that require an alert
Abstraction	○ Relevant Information: Peak Signal Amplitude (the "Metal Magnitude") and the current baseline signal. ○ Irrelevant Information: The shape of the object (Spatial Density), the type of metal (Phase Angle), and how long it stays in the gate (Signal Duration).

ITERATION 2 - Conductivity Checking	
Problem Identification	How can we automate the detection of inductive metal items to identify suspicious objects like vapes/e-cigarettes, considering that some objects have a plastic or non-metal casing?
Decomposition	● Signature Extraction: How do we isolate the specific Phase Angle of the lithium-ion batteries and heating coils found in vapes? ● Material Classification: How do we differentiate between high-conductivity metals (aluminum, lithium, copper) and low-conductivity metals (iron, steel)? ● Casing Penetration: How do we ensure the sensor ignores the plastic exterior and focuses on the internal conductive components?
Pattern Recognition	● The "Electronic Fingerprint": Vapes produce a

	specific phase shift that differs from "safe" items like a soda can or a foil gum wrapper. • Conductivity vs. Amplitude: Recognizing that a vape may have a small Signal Amplitude (small size) but a very high Phase Angle (high conductivity).
Abstraction	• Relevant Information: Phase Angle (Conductivity) and Inductive Response. • Irrelevant Information: The shape, color, or material of the outer casing (plastic, rubber, or leather).

ITERATION 3 - Spatial Density Checking	
Problem Identification	How can we differentiate between a single large, high-threat object and multiple small, harmless items that create a large total signal? Multiple small items often appear as one large object to standard sensors, leading to unnecessary manual bag checks and bottlenecks.
Decomposition	• Spatial Mapping: How do we map the spatial_density across the sensor array to see where the metal is physically concentrated? • Peak Isolation: How do we distinguish between one sharp peak in density versus multiple smaller, distributed peaks? • Threshold Balancing: How do we set a density level that ignores a cluster of keys but flags a solid metallic plate or tool?
Pattern Recognition	• The "Solid Mass" Signature: Characterized by high spatial_density (Levels 7–10) concentrated in a single geometric coordinate. • The "Bag Clutter" Pattern: Characterized by low-to-medium spatial_density (Levels 1–6) spread across a wider area, indicating multiple distinct items like pens, coins, and keys. • Density vs. Magnitude: Recognizing that a bag of change might have the same "Metal Magnitude" as a weapon, but the density "shape" will be scattered and uneven.
Abstraction	• Relevant Information: spatial_density (concentration of metal) and the location of signal peaks. • Irrelevant Information: The total weight of the

	bag, the number of non-metallic items, and the specific brand of the bag.
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Techniques Used in Solving the Problem	
Encapsulation	Created Bag and MetalDetector classes to manage data.
Solution Modeling	Developed a scoring formula based on metal, duration, angle, and density.
Optimization	Used Pandas to handle large datasets efficiently.
Dynamic Thresholding	Categorized threat levels (LOW, MEDIUM, HIGH) based on score averages.

Source Code
<pre>import pandas as pd import random data = pd.read_csv(r'/content/drive/MyDrive/Students_bag_dataset.csv') class Bag: def __init__(self, bag_id, data): self.id = bag_id self.items = [] for _, row in data[data['Bag_ID'] == bag_id].iterrows(): self.items.append({ 'name': row['Object_Name'], 'metal': 1 if row['Metal_Presence'] == 'Yes' else 0, 'duration': random.randint(1, 10), 'angle': random.randint(1, 10), 'density': random.randint(1, 10), 'sharp': row['Sharp_Object'] == 'Yes' }) class MetalDetector: def scan(self, bag): score = 0 for item in bag.items:</pre>

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        score += item['metal'] * 10
        score += item['duration']
        score += item['angle']
        score += item['density']
        if item['sharp']:
            score += 30

    avg = score / len(bag.items) if bag.items else 0
    return "HIGH" if avg > 25 else "MEDIUM" if avg > 15 else "LOW"

all_bag_ids = data['Bag_ID'].unique()

detector = MetalDetector()

print("BAG SECURITY SCAN RESULTS")
print("=" * 40)

for bag_id in all_bag_ids:
    bag = Bag(bag_id, data)

    threat_level = detector.scan(bag)

    dangerous_count = sum(1 for item in bag.items if item['sharp'])

    print(f"{bag_id}: {threat_level} threat", end="")
    if dangerous_count > 0:
        print(f" ({dangerous_count} sharp objects!!)")
    else:
        print()

print("=" * 40)
print(f"Total bags scanned: {len(all_bag_ids)}")

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Sample Output

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... BAG SECURITY SCAN RESULTS
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BAG_001: MEDIUM threat
BAG_002: HIGH threat (2 sharp objects!!)
BAG_003: HIGH threat
BAG_004: MEDIUM threat
BAG_005: HIGH threat (2 sharp objects!!)
BAG_006: HIGH threat (3 sharp objects!!)
BAG_007: HIGH threat
BAG_008: HIGH threat (2 sharp objects!!)
BAG_009: HIGH threat (2 sharp objects!!)
BAG_010: HIGH threat
BAG_011: HIGH threat
BAG_012: HIGH threat (1 sharp objects!!)
BAG_013: HIGH threat (1 sharp objects!!)
BAG_014: HIGH threat (2 sharp objects!!)
BAG_015: HIGH threat (2 sharp objects!!)
BAG_016: MEDIUM threat
BAG_017: HIGH threat (1 sharp objects!!)
BAG_018: MEDIUM threat
BAG_019: HIGH threat (2 sharp objects!!)
BAG_020: HIGH threat
BAG_021: HIGH threat (2 sharp objects!!)
BAG_022: HIGH threat
BAG_023: HIGH threat (2 sharp objects!!)
BAG_024: MEDIUM threat (1 sharp objects!!)
BAG_025: MEDIUM threat
BAG_026: HIGH threat (1 sharp objects!!)
BAG_027: MEDIUM threat
BAG_028: MEDIUM threat
BAG_029: HIGH threat (1 sharp objects!!)
BAG_030: MEDIUM threat
=====
Total bags scanned: 30
```

Google Colab Link

<https://colab.research.google.com/drive/1rnsDH6JWadDwsMy-DzdoZ6NitfV3Db0y?usp=sharing>

Github Link

<https://github.com/qrbvamora-source/Case-Study-1.git>

Conclusion

In this case study, we developed a Bag Security Scan System to automate threat detection at school entrances, improving safety and reducing congestion. The hardest parts of the project included generating realistic sensor data using ChatGPT to create the synthetic CSV dataset, simulating complex real-world signals like duration and density, and tuning the scoring logic to accurately classify threat levels without physical hardware.

We used object-oriented design with Bag and MetalDetector classes to keep the system scalable and clear. By decomposing the problem into three iterations—Metal Magnitude, Conductivity, and Spatial Density—we were able to address the "clutter effect" that often plagues traditional security systems.

The final program successfully scans bags, flags sharp objects, and outputs actionable threat levels, proving that a multi-sensor classification system can be modeled effectively through careful data simulation and logical scoring—laying the groundwork for a future real-time, automated security solution that eliminates manual bottlenecks.

Members

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